

Nahid Hasan Sagar

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PROFESSIONAL SUMMARY

M.S. candidate in Statistical Science at Indiana University Bloomington with a B.S. in Physics and experience developing quantitative research software in Python, R, and C++. Applies Bayesian modeling, statistical learning, time-series analysis, and numerical methods to data-intensive problems, with a focus on rigorous validation, reproducibility, and computational performance.

EDUCATION

Indiana University, College of Arts and Sciences

Master of Science in Statistical Science

Thesis: Time-Series Analysis Using Bayesian Additive Regression Trees (BART)

Bloomington, IN

Expected May 2027

Relevant Coursework: Introduction to Statistical Computing; Applied Statistical Computing; Fundamentals of Statistics I–II; Applied Linear Models I–II; Statistical Consulting; Research in Statistics; Introduction to Analysis I

University of Dhaka

Bachelor of Science in Physics

Dhaka, Bangladesh

August 2024

RESEARCH EXPERIENCE

Research Assistant, StatMIND Lab

Indiana University Bloomington

2025 – Present

Bloomington, IN

- Develop Bayesian brain-mapping algorithms for high-dimensional neuroimaging data, translating statistical research methods into efficient C++ implementations integrated with R through Rcpp and RcppArmadillo.
- Reproduce results from the original R package and validate the new implementation with unit tests, benchmark comparisons, and systematic numerical checks.
- Achieve a $4\text{--}5\times$ speedup in core computations, enabling faster iteration on large-scale datasets.
- Build reproducible workflows to compare runtime, scalability, and computational performance across implementations.

SELECTED RESEARCH & PROJECTS

M.S. Thesis: Time-Series Analysis Using BART

R, Bayesian Modeling

- Conduct research on Bayesian Additive Regression Trees for time-series analysis under the supervision of Dr. Matthew T. Pratola, evaluating model behavior for structured temporal data.

Ordinary Least Squares from Scratch

Python, NumPy, Pandas, Matplotlib

- Implement ordinary least squares regression from first principles using matrix-based estimation, with transparent computation and data-analysis visualizations.

TECHNICAL SKILLS

Programming: Python, R, C++, C, MATLAB, Bash

Quantitative Methods: Bayesian inference, BART, MCMC, time-series analysis, linear and logistic regression, mixed-effects models, bootstrap methods, numerical optimization, model selection, cross-validation, classification metrics, predictive model evaluation

Research Engineering: Algorithm benchmarking, code profiling, unit testing, reproducible workflows, Git, GitHub, LaTeX, RStudio, VS Code

LEADERSHIP & HONORS

- Attended the 2026 Velocity Conference at UC Berkeley as an Indiana University representative for the science community.
- Co-organized the Australian Mathematics Competition in Bangladesh in 2014.
- Taught and mentored more than 100,000 students through classroom instruction and national online learning platforms as a Senior Mathematics Instructor.